INORGANIC CUME
Oct-10-2008

(1). A trace amount of CO in the H_2 stream can poison the hydrogen fuel cell catalyst that is usually the Pt nanocrystal. Please use your inorganic chemistry knowledge to explain such poisoning process. (10')

(2). A trace amount of sulfur-containing molecules in the petroleum-based fuel (reformat) can poison the Pt catalyst in the reformat fuel cell. Please use your inorganic chemistry knowledge to explain this catalyst deactivation. (10')

(3). If the Pt-catalyst is a face center cubic crystal with the diameter around 10 nanometers, which spot(s) on the nano-cube would be poisoned most easily and which the most difficultly, relatively? Give your reason(s). (20')

(4). For regenerating the catalysts, heating for removing the CO or the sulfur-molecules may end up with the in crease of the catalyst-particle size thus the much less active catalyst. Propose a way to potentially remove both poisoning species from the catalyst surface with reducing the catalyst's activity. (20')

(5). Propose a way to remove the CO from the H2, and propose another method for removing the sulfur-molecule from the reformat. (20')